TWMailer Basic - Protol

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1. Client and Server Architecture

TW-Mailer uses a client-server architecture based on TCP sockets. The client provides a command-line interface for sending, listing, reading and deleting messages. The server processes incoming requests and stores messages in user-specific inbox directories.

The server operates iteratively, handling one client at a time. Multiple commands can be executed over a single connection until `QUIT` is sent.

Communication uses a newline-delimited text protocol with five commands: `SEND`, `LIST`, `READ`, `DEL` and `QUIT`. Multi-line message bodies are terminated by a line containing `.`.

2. Used Technologies

- C (C11): Implementation language.
- POSIX/Linux: Filesystem operations, dynamic memory management and system calls.
- TCP/IP sockets: Client-server communication using the BSD socket API.
- Make and CMake: Building and managing the client and server.

3. Development Strategy and Adaptations

Because TCP is a byte stream without message boundaries, newline-based framing was implemented. Dynamic buffers handle variable-length input, while dedicated send functions account for partial transmissions.

The server stores each message as a separate `.msg` file inside the recipient's inbox. Message IDs determine their ordering. Input validation, error handling and temporary files for message storage were implemented to improve reliability.